JAWS: JGI Analysis Workflow Service

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JGI JOINT GENOME INSTITUTE

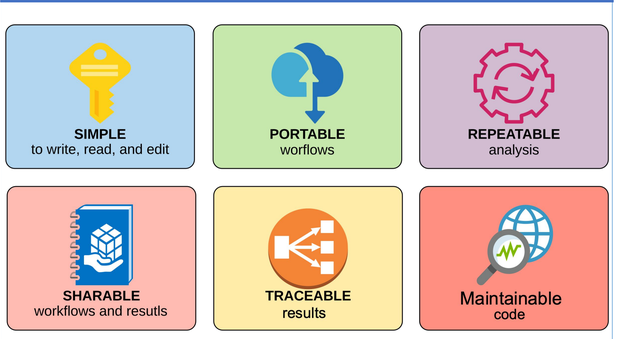
Introduction

Most scientific analysis workflows are

- Written in different programming languages with different structures with complicated logic
- Using Idiosyncratic glue code to handle job submission, checkpointing, logging, etc
- Customized to specific compute environments

This has been making it difficult to improve **maintainability, portability, and shareability** of the workflows.

Our Goals

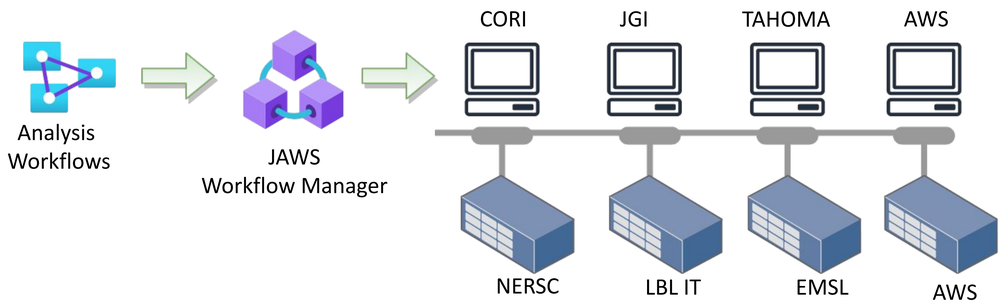


Our Solution

- The Joint Genome Institute (JGI) has developed **JGI Analysis Workflow Service (JAWS)** as a framework to run computational workflows
- The purpose of JAWS is to improve the **reusability** and **robustness** of bioinformatic (or any) workflows in evolving and/or diverse high-performance computing (HPC) and cloud environments
- JAWS automatically transfers data and code to user-specified compute resources, executes computation, and returns result files
- JAWS uses **Cromwell**, developed at the Broad Institute, to execute workflows in **Workflow Description Language (WDL)**
- JAWS supports various container techs, such as Docker, Shifter, and Singularity to ensure the **portability** and **maintainability**
- **HTCondor** serves as the backend to Cromwell and handles the running of the jobs on a HPC cluster

What is JAWS?

JAWS is a user-friendly workflow manager to seamlessly route jobs and data across multiple sites



A common interface to access resources makes distributing analysis and data transparent to the user

JAWS Components and Architecture

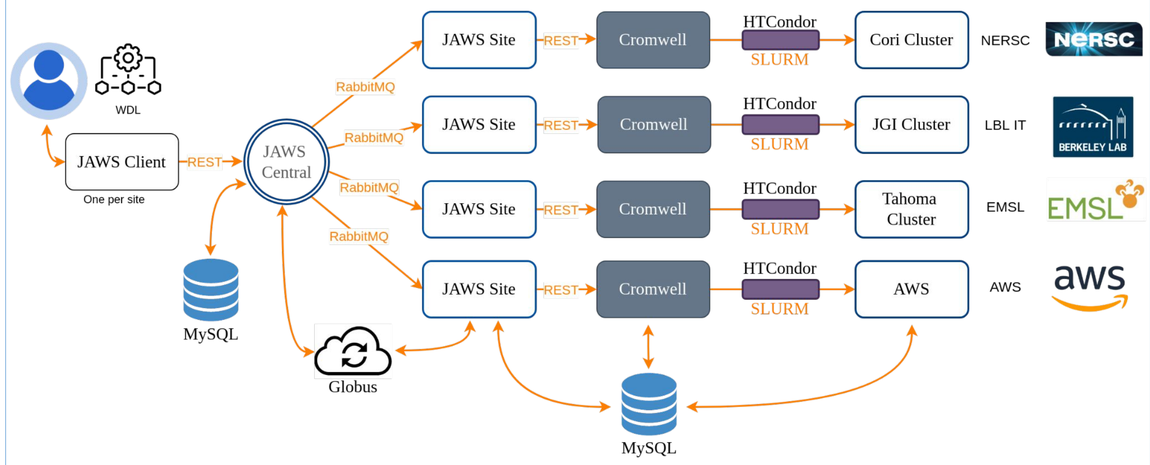
- **Client**
  User-friendly command line interface with rich features
- **Central**
  Dispatches jobs to sites and tracks user history
- **Site**
  Extends Cromwell with additional features
- **Cromwell**
  Workflow execution engine
- **HTCondor**
  Flexible and efficient JAWS backend for SLURM

JAWS JGI Analysis Workflow Service

{wdl} Workflow Description Language

Cromwell

docker



JAWS Main Features

Features	Cromwell	JAWS
WDL linter (validation)	✔	✔
Run WDL/CWL workflows	✔	✔
Configurable compute resource	✔	✔
Data transfer (using Globus)		✔
Multiple cluster support		✔
Cromwell-backend middleware		✔
User account, history, logging		✔

JAWS Ecosystem

- **Dashboard**
  A graphical user-interface for users
- **Monitor**
  Dashboard of system health
- **Centralized logging**
  Logs for all services/sites can be found in one place that also supports elastic search
- **Automated end-to-end testing**
  Process for quickly identifying issues with underlying infrastructure
- **Workflow repositories**
  Workflows created by any JGI Department are registered and shared
- **Subworkflow catalog and Docker hub**
  Reusable building blocks for building new workflows

Compute Resource Supported

Total 3,088 compute nodes from the clusters at NERSC, LBL IT, EMSL, and AWS

Site	Pool	Nodes	Memory(G)	Cores
CORI NERSC	Medium	2,388	118	64
	Xlarge	20	1,450	72
JGI Lab-IT	Smal	316	46	64
	Medium	72	236	64
	Large	8	492	64
Tahoma EMSL	Medium	160	364	72
	XLarge	24	1,480	72
AWS	-	100	512	64

Future Plans

- Training for wider adoption of WDL and JAWS
- Intelligent dispatch of workloads to compute sites
- Resource usage monitoring and reporting
- Additional compute sites: ORNL, AWS, Google Cloud

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The work conducted by the Joint Genome Institute is supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.